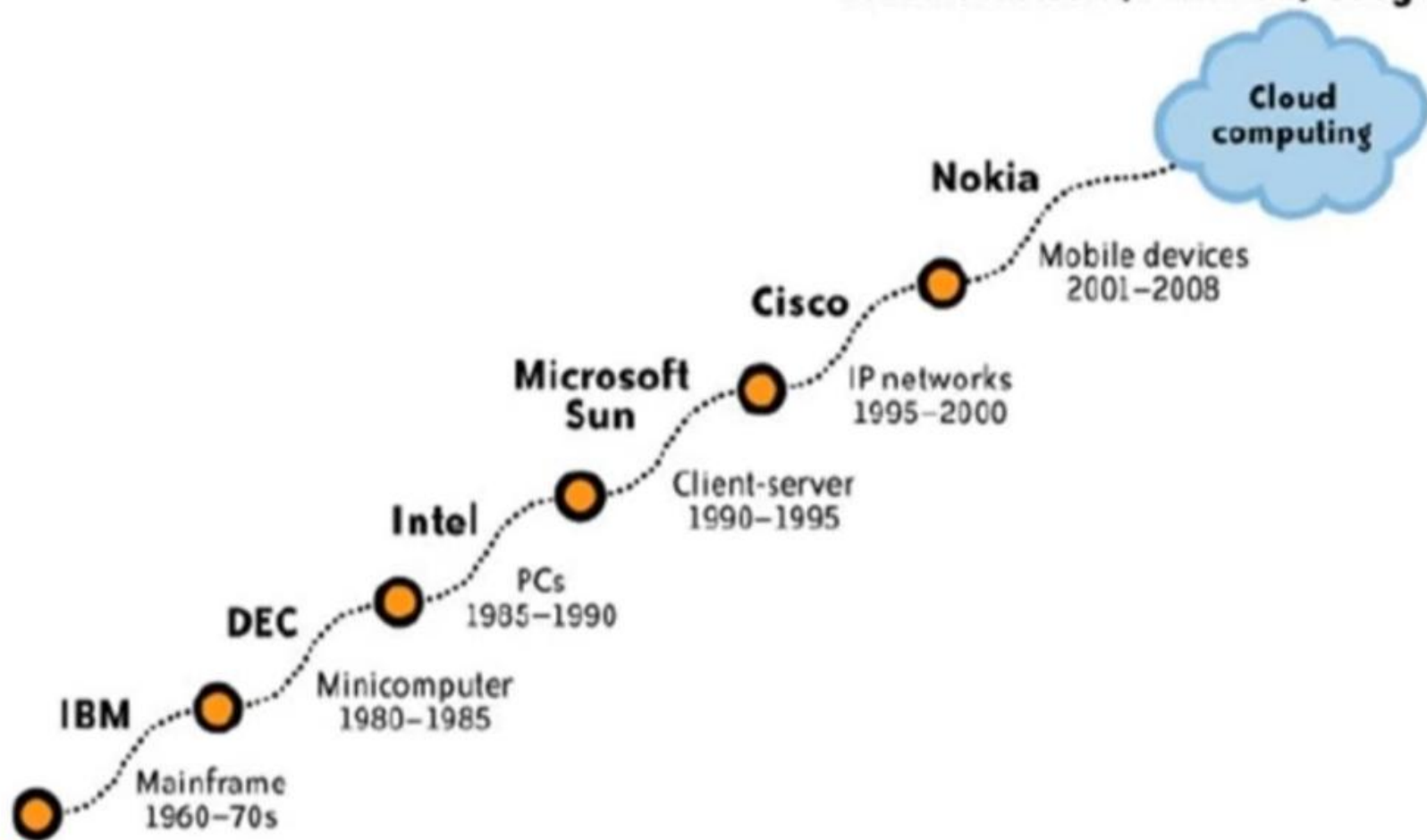






On-Premises Servers

- Bring your own machines, connectivity, software, etc.
- Complete control
- Complete responsibility
- Static capabilities
- Upfront capital costs for the infrastructure



Hosted Servers

- Renting machines, connectivity, software
- Less control
- Fewer responsibilities
- Lower capital costs
- More flexible
- Pay for fixed capacity, even if idle



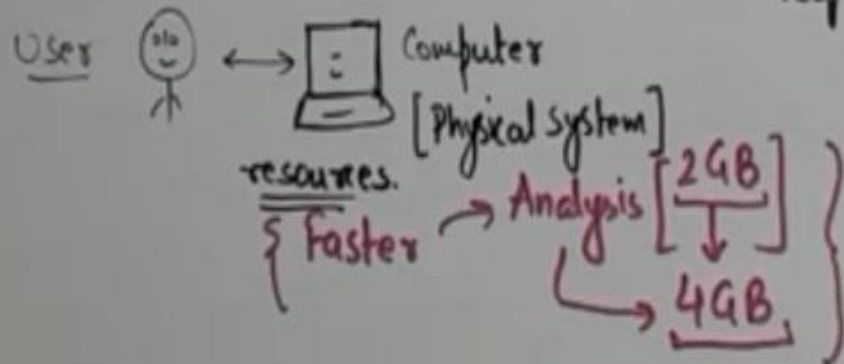
Cloud Platform

- Shared, multi-tenant infrastructure
- Virtualized & dynamic
- Scalable & available
- Abstracted from the infrastructure
- Higher-level services
- Pay as you go

Cloud Computing:- It provides us the means by which we can access the applications as utilities over the internet.

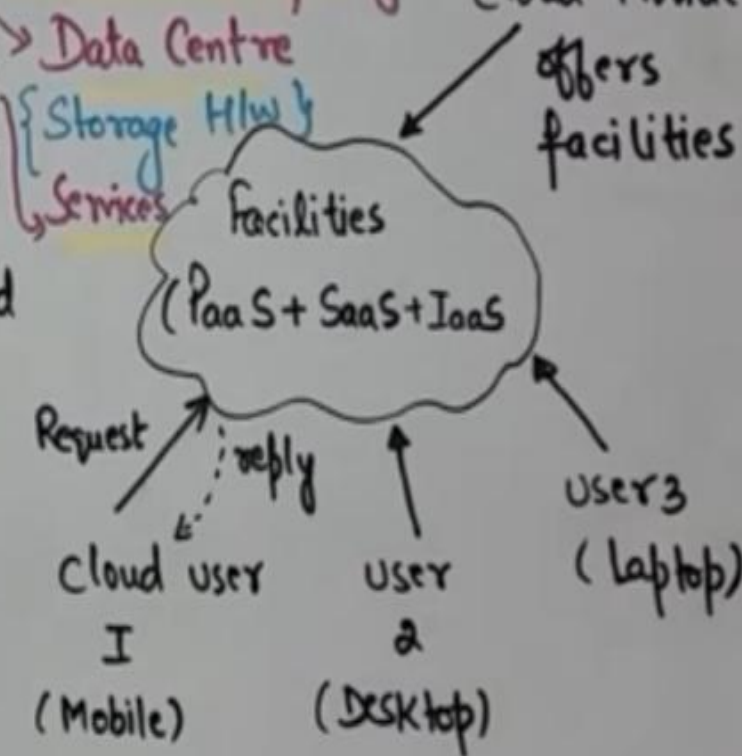
CLOUD: It is something which is present at remote location.

Cloud Computing refers to manipulating, configuring, and accessing the applⁿ online. as per req.



Basic Structure: → data can be accessed from anywhere.

→ off-premise computing. Cloud Provider



Characteristic of Cloud Computing: → Deployment models, Service models, Essential and common characteristics.

On-demand Self-Service: Users can use the cloud services as per demand.

Broad N/W Access: Can be accessed anytime from anywhere.

Resource Pooling: multiple users sharing pool of resources

Scalability: Resources can be scaled at any time.

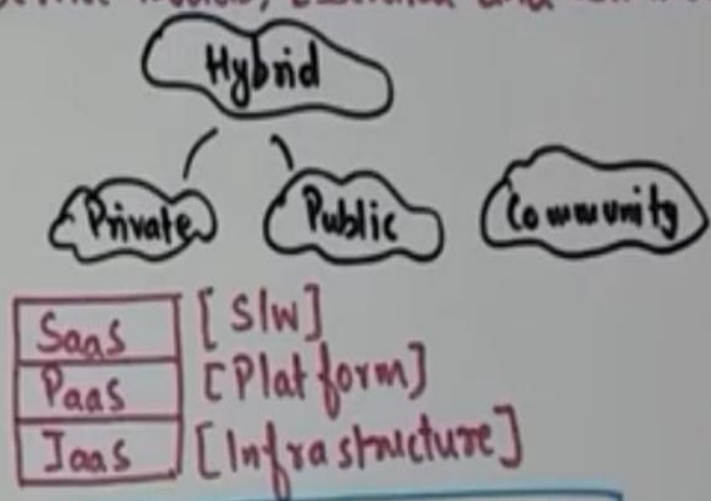
Measured Service: Monitoring tool

Deployment models

Service models

Essential characteristics

Common ch.



On-demand Self-Service
Broad n/w access.

Virtualization, Geographical distrib.,
Low cost Applⁿ, Advanced security,
Service Oriented, Massive scale.

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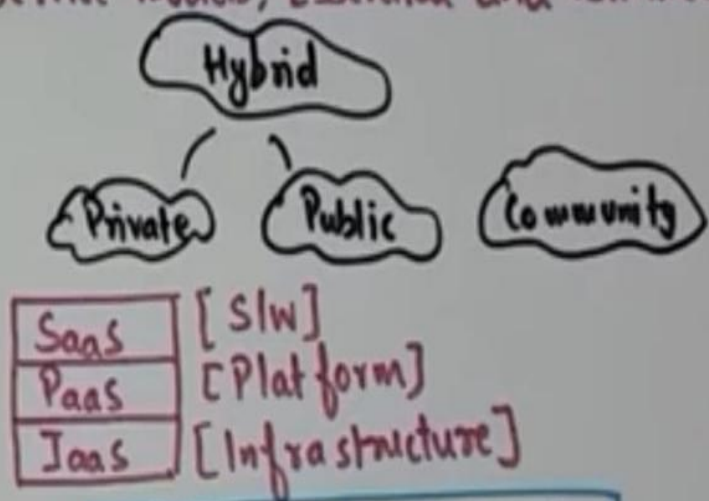
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Service models

Essential characteristics

Common ch.



On-demand Self-Service
Broad n/w access.

Virtualization, Geographical distrib.,
Low cost Applⁿ, Advanced security,
Service Oriented, Massive scale.

Working Models for cloud Computing:

(i) Deployment Models: It defines the type of access to the cloud.

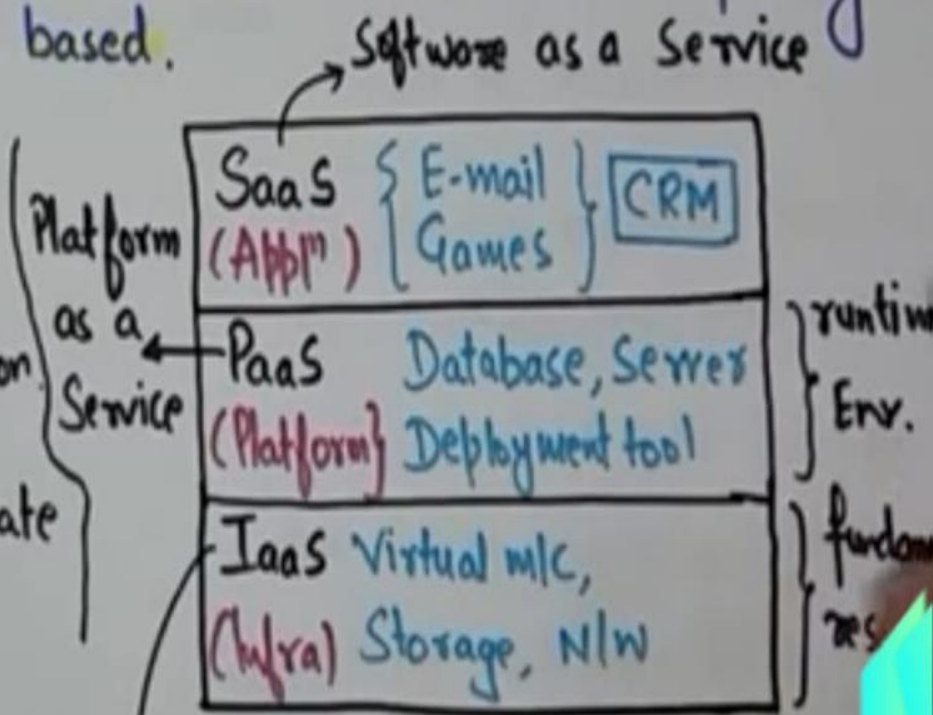
↳ Public :- Services are accessible to the general public. [Less Secure].

Private :- Accessible within an organization. [More Secure]

Hybrid :- Public + Private Critical → Private
Non-Critical → Public.

Community :- Accessible by Group

(iii) Service Models: These are the reference models on which the cloud computing is based.



CLOUD MODEL-Types:-

(i) PUBLIC CLOUD MODEL: { Small fee, cost, free }

↳ Systems and Services are easily accessible to general public.

Ex:-
→ GOOGLE
→ AMAZON
→ MICROSOFT } Internet.

Advantages:-

- i) Reliability
- ii) Flexibility
- iii) Location independence
- iv) High scalability
- v) Cost effective
- vi) utility style costing.

Disadvantages:-

- (i) Low Security
- (ii) Less customizable than private cloud.



Public cloud model.

Free
Gmail (15GB)

↓ >15GB

Fee

(iii) PRIVATE CLOUD MODEL: { managed by third-party also }

↳ Systems and Services are accessible only within the organization.

[Microsoft, VMware, Elastic-private cloud, ubuntu]

Advantages:-

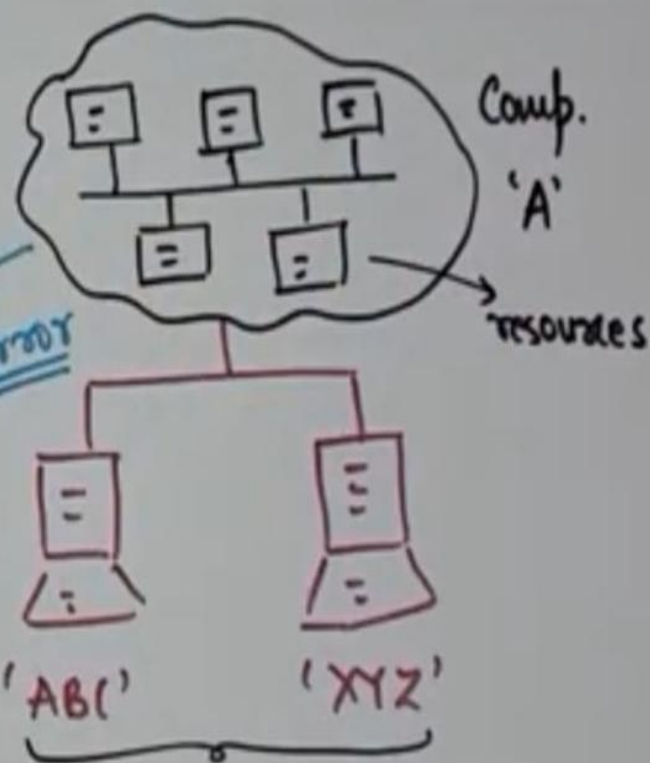
- (i) High Security and privacy
- (ii) More Control
- (iii) Cost and Energy efficient
- (iv) Improved Reliability

is used only by trusted people.

Disadvantages:-

- ↳ i) Area of opⁿ is limited
- ii) Price is ↑.
- iii) Scalability is limited
- iv) Skilled people

(M) Users * Error



Both the users should be member of Comp. 'A'

HYBRID CLOUD MODEL:

↳ Combination of both PUBLIC and PRIVATE Cloud.

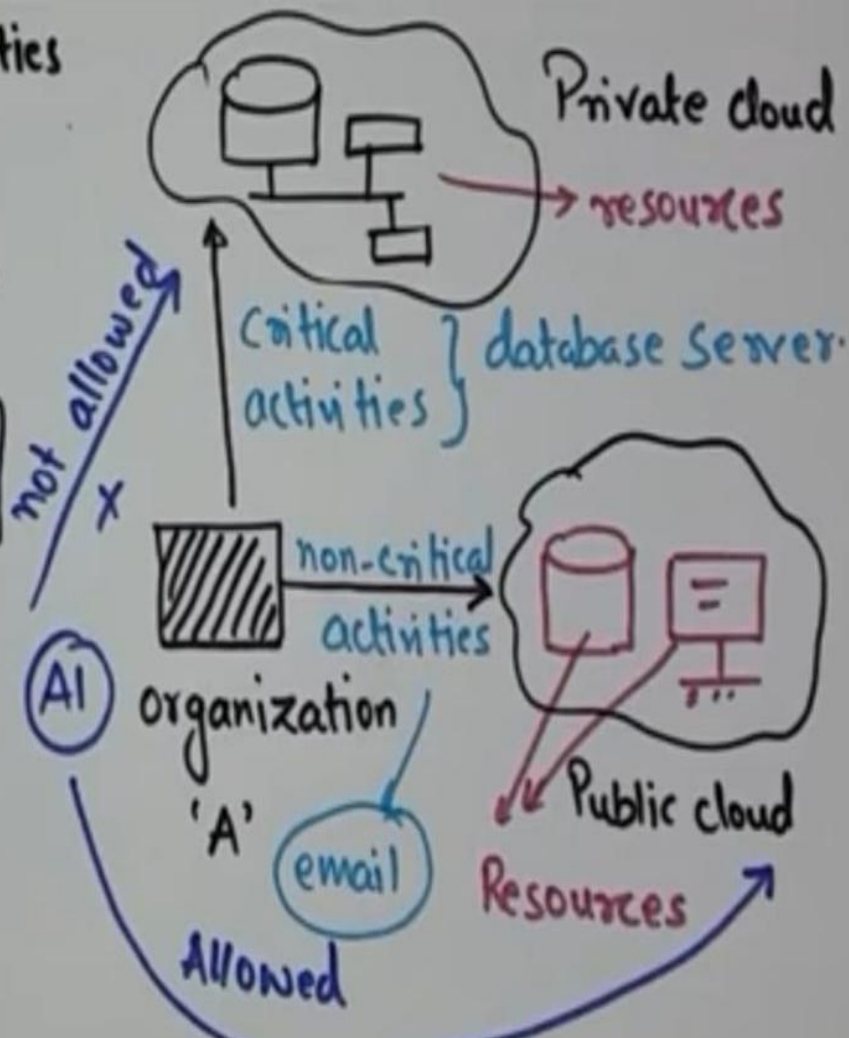
non-critical activities
critical activities

Advantages:

- ↳ i) Scalability
- ii) Security
- iii) Flexibility
- iv) ↓ Cost

Disadvantages:

- ↳ i) N/w Issues [Complexity of both]
 - ii) Security Compliance
 - iii) Dependency on the Infrastructure.
- Internal IT.



Community cloud Model: falls b/w category of private and public cloud.

↳ Allows systems and services to be accessible by group of organizations. } COSTLY than public cloud.

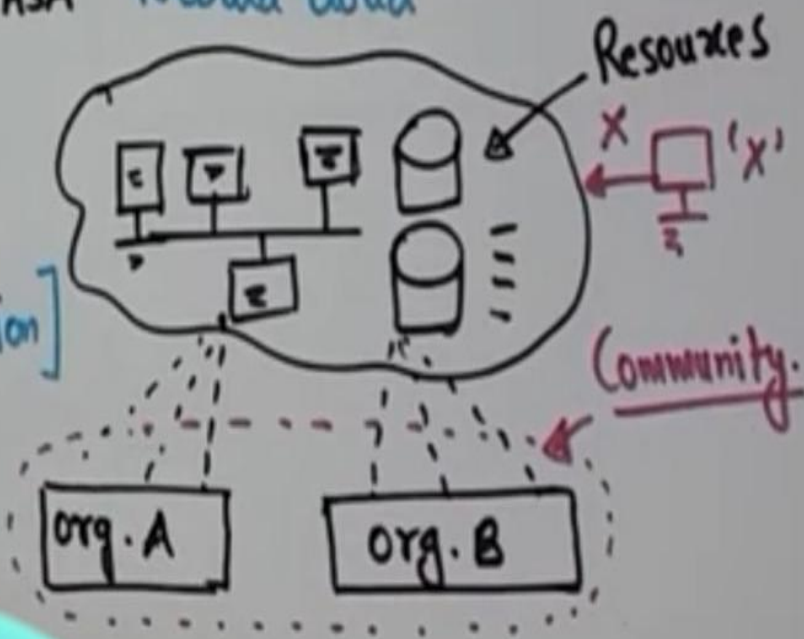
→ Google 'Gov cloud'
→ NASA 'Nebula cloud'

Advantages:

- ↳ i) Low cost than private cloud.
- ii) Sharing among org.
- iii) Security [more secure than public but less than private]

Disadvantages:

↳ we need to be careful in storing data [data segregation]



Software-as-a-Service: (SaaS)

↳ offers S/W applⁿ as a Service to the users.

Applications: → (Sales Force)

↳ i) CRM applⁿ. [Customer relⁿ mgmt]

ii) Help Desk

iii) HR Solⁿ

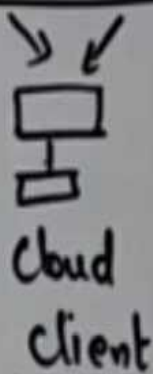
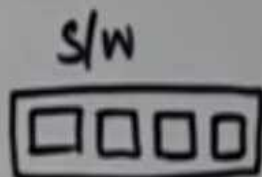
iv) Billing and Invoicing.

Cloud Provider



Physical resource

Virtual resource



Client-Server technology.

May Pay low cost to use the S/W.

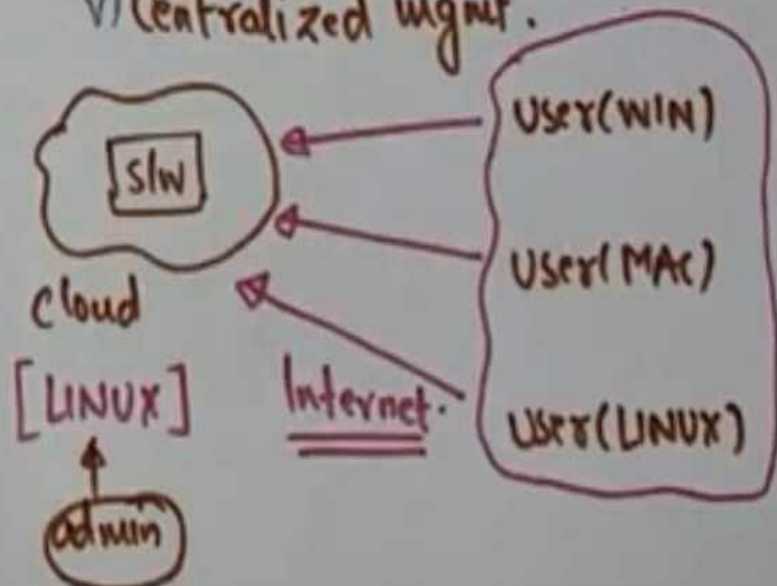
Characteristics of SAAS:

- Availability of S/W over internet.
- Maintenance of S/W by Vendor. } upgrade
- Subscription or Usage based license.
- COST effective } Pay as per use.
- On-demand availability. } anywhere, anytime
- Easily Scalable as per need.
- Works on Shared model. } client 1, 2, 3
- Automatic updation of S/W.

→ Client will always gets to work on -the latest version of the S/W.

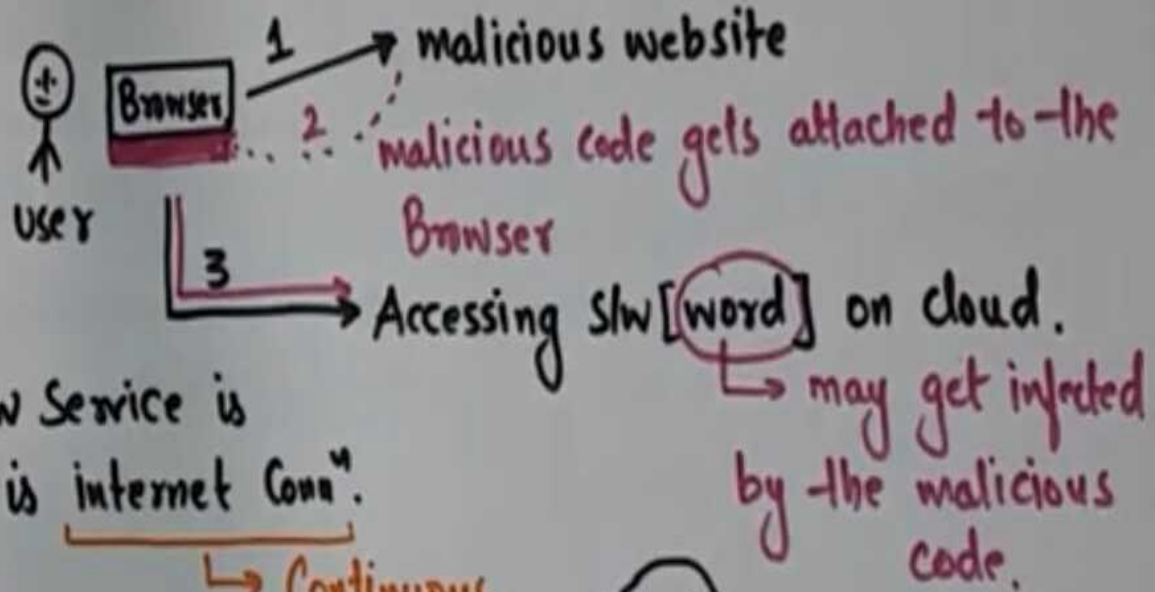
Benefits of SAAS:

- i) Modern S/W tools.
- ii) Platform independence to user.
- iii) Efficient use of S/W License.
- iv) Multitenant serⁿ.
- v) Centralized mgmt.



SAAS ISSUES:-

→ i) Browser based risks

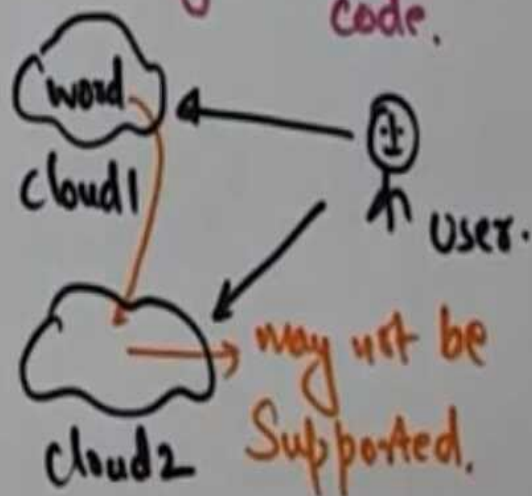


ii) N/w dependence: S/W Service is available till there is internet Connⁿ.

→ Continuous.

iii) Portability Issues.

→ among diff. SaaS clouds.



PAAS (Platform-as-a Service) Platform $\Rightarrow$ Computing platform [HW, OS, Libraries]

$\rightarrow$ It offers runtime environment for the applications.

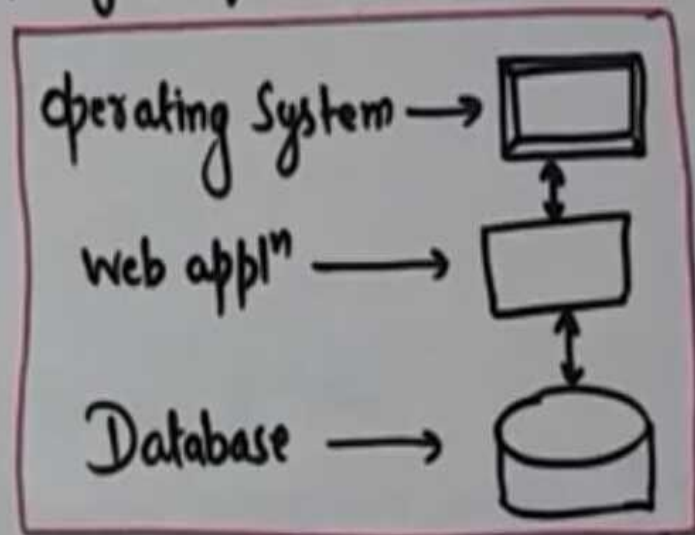
$\rightarrow$ Basically provide platform to develop applⁿ.

Eg:- Google App Engine } Helping the user
Force.com } to create web
applⁿ online.



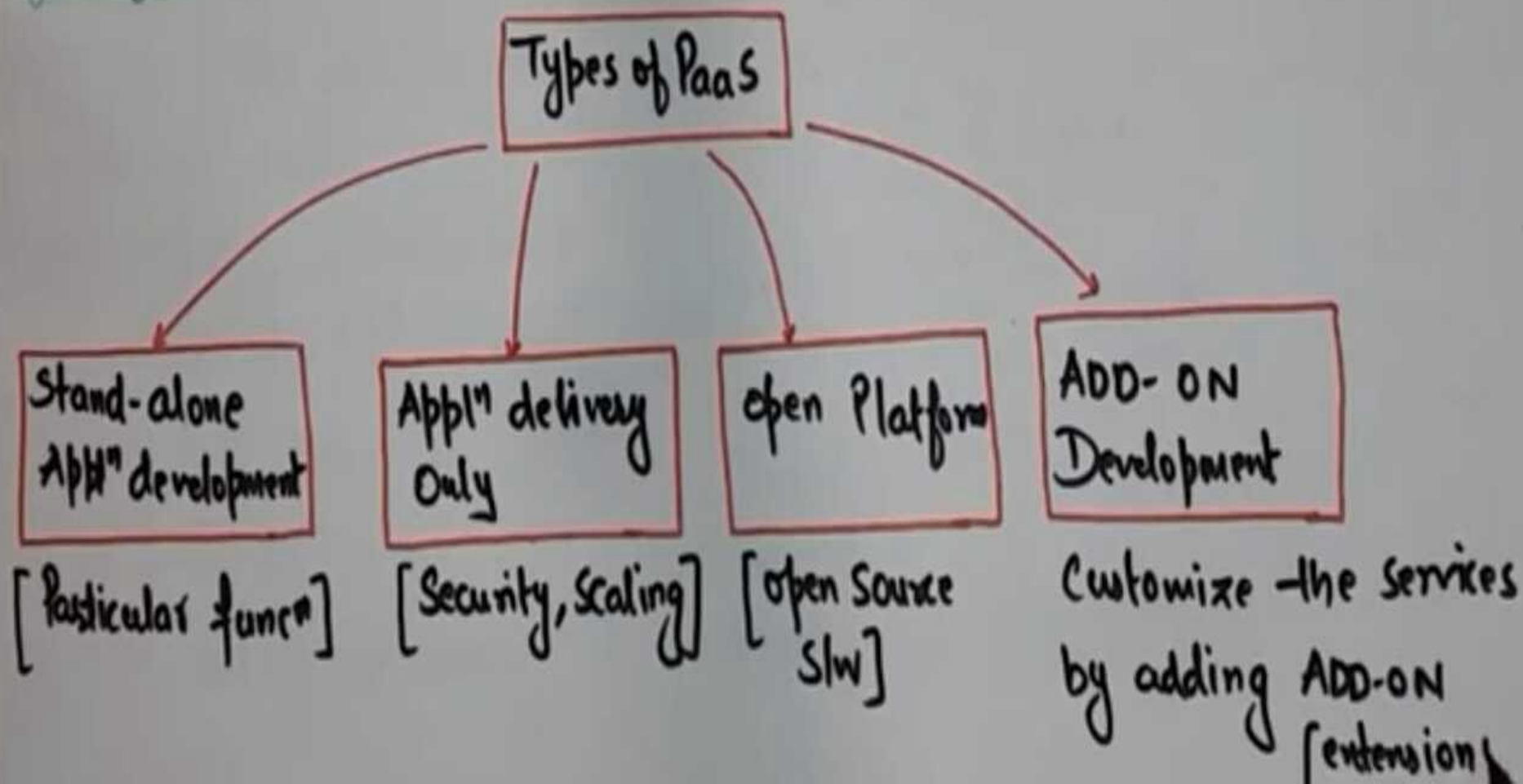
USER

(LOG IN)
Coding.



Paas (cloud providers)

Types of PAAS:-



Benefits of PAAS:

- i) Lower COST] Customer doesn't need to purchase H/W, S/W.
- ii) SCALABILITY] resources can be scale up & scale down easily.
- iii) UPdated S/W System [S/W maintenance is - the issue/res. of vendor]
- iv) Less admin. overhead.
→ (administration is done by cloud provider)

Issues of PAAS:

- i) Portability issue among PaaS cloud providers.
- ii) Event based processor scheduling.
- iii) Security. [latest cryptographic ALGO should be used]

Characteristics of PAAS:

- i) Browser based environment for the development of applⁿ.
- ii) Secured, Scalable web services
- iii) Easy workflow & Approval process.
- iv) Easy Integration with other applications.

IAAS [Infrastructure-as-a Service]

↳ Provides access to

- Physical machines
- Virtual machines
- Virtual Storage.

operates by Hypervisor (KVM or Xen)

↳ other things that offers:

↳ i) Virtual LAN (VLANs)] N/w device (Physical resources)

ii) IP addresses

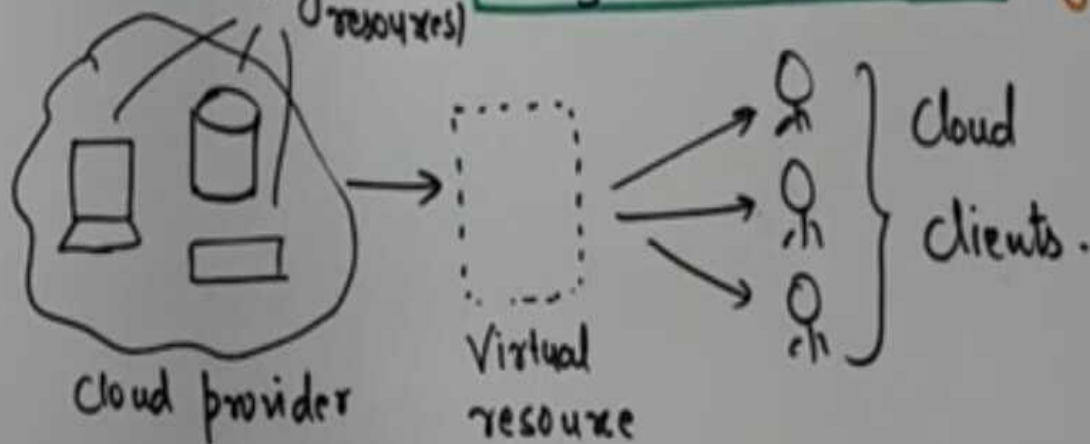
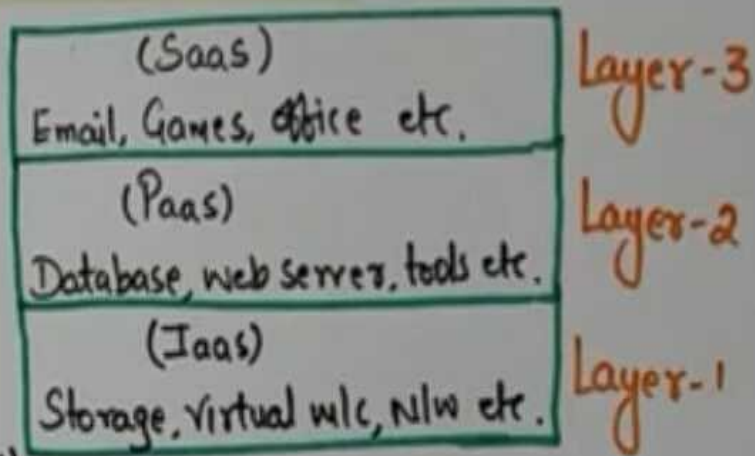
iii) SW bundles

iv) Load balancers

v) Disk storage

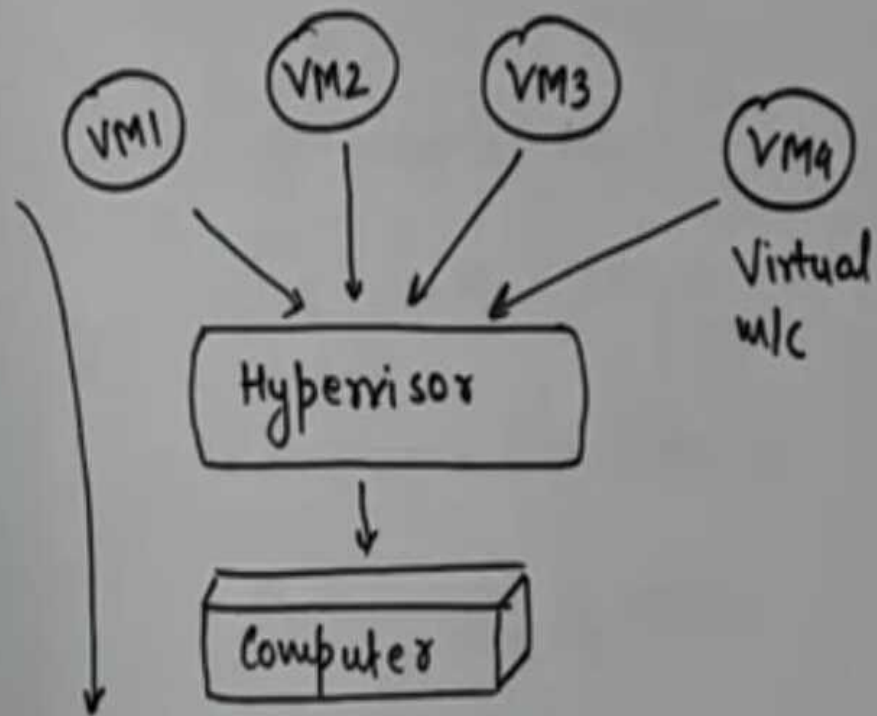
[AMAZON, Google Compute Engine]

cloud Service Models:



Benefits of IAAS: Using IAAS, cloud provider can freely locate infrastructure over the internet in cost-effective manner.

- ↳ i) Global Accessibility
ii) Easy Integration of devices
iii) Scalability is Easy
iv) Availability is high
v) Flexibility
vi) Full control to Virtual m/c.



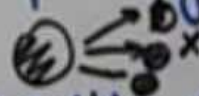
Services is being provided by Virtual m/c through hypervisor.

IAAS ISSUES:-

- ↳ i) Compatibility:- Consumer need to use only legacy SW on the Infrastructure.
 - ↳ ↑ COST.
- ii) Virtual Machine:- Mismatch b/w the VM security and cloud provide Security Version.
- iii) Robustness:- Splitting of Single system resource into multiple virtual machines.
- iv) Data Deletion:- if client deletes the data - then provider should also delete it permanently.

Characteristics of IAAS:-

- ↳ i) Pre-Installed SW.
 - ↳ on virtual m/c.
- ✓ ii) Pre-Installed OS.
 - ↳ window, linux, Solaris
- ✓ iii) Resources On-demand
 - ↳ available.
- ✓ iv) Multiple copy of data
- ✓ v) Scalability of computing.



Characteristic of Cloud Computing: → Deployment models, Service models, Essential and common characteristics.

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Broad N/W Access: Can be accessed anytime from anywhere.

Resource Pooling: multiple users sharing pool of resources

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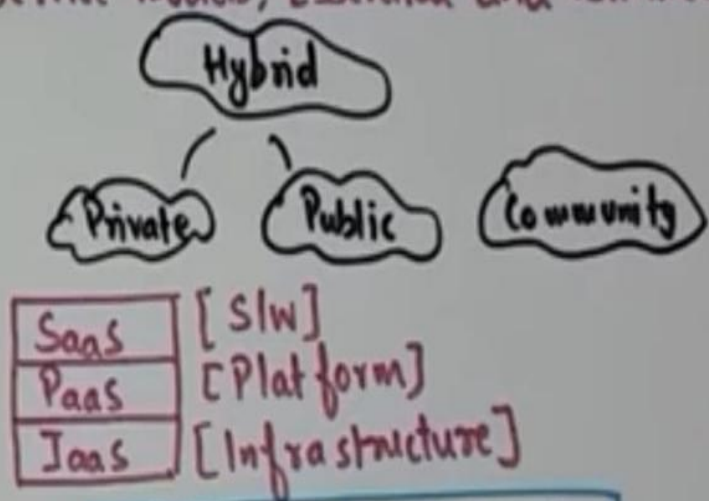
Measured Service: Monitoring tool

Deployment models

Service models

Essential characteristics

Common ch.



On-demand Self-Service
Broad n/w access

Virtualization, Geographical distrib.,
Low cost Applⁿ, Advanced security,
Service Oriented, Massive scale.

CLOUD COMPUTING PROVIDERS:

① Amazon Web Services (AWS):

↳ Provides services such as computation power, storage etc.

↳ flexible, reliable and scalable

↳ (Pay per usage.)

② Microsoft Azure: used for building, testing, deploying and managing applⁿ.

↳ uses Hypervisor } → Virtualization

③ Google cloud: Basically used for Google Search and Youtube.

↳ Services offered:
↳ Data Analysis
↳ Machine Learning
↳ Storage.
↳ Apps.

④ Salesforce.Com: Development Platform. [CRM]

IBM

↳ Customer relⁿ management

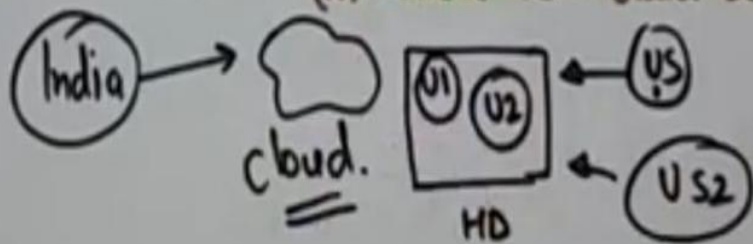
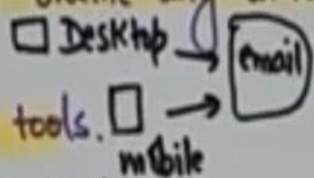
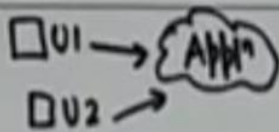
CLOUD COMPUTING Benefits:

- i) Access applⁿ over the internet.
- ii) Manipulation and Configuration of the applⁿ online any time.
- iii) No SW installation is required.
- iv) PAAS: Online development and deployment tools.
- v) On-demand Self service.
- vi) Cost Effective.
- vii) More Reliable.
- viii) Efficient Load Balancing.
- ix) Fast access of resources due to parallel computing.

Resources are allocated as per User need.

Internet Connⁿ.

Load balancers.



RISKS of cloud Computing:

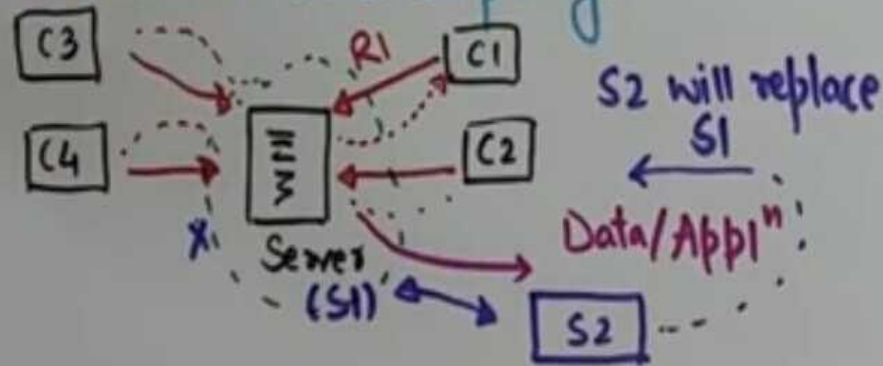
- (i) By whom data and applⁿ will be accessed.
[Scrutiny]
- (ii) Security methods for the data storage and transmission
[Secured Pwd]
- (iii) How data & Applⁿ from various consumers reserved separately.
- (iv) where will data be stored.

History of cloud Computing:

- 1999 (Salesforce.com, Pioneer)
- 2002 (Amazon → AWS (Amazon Web Services) (Customized cloud Services))
- 2009 (Google → Google Apps)

Client-Server Technology:

- It is behind cloud computing.



Advantages of client-Server:

- i) Centralized Control [Easy to manage]
- ii) Managed by trusted admin.
- iii) Backup is there for replication
- iv) Easy to update
- v) Remote Services.
- vi) [Secure and Safe]



Disadvantages:

- i) Single Server] Congestion → unable to handle multiple request.
- ii) Download may get discontinued due to some failure.
- iii) Management is costly.
- iv) Expert IT Services is required to

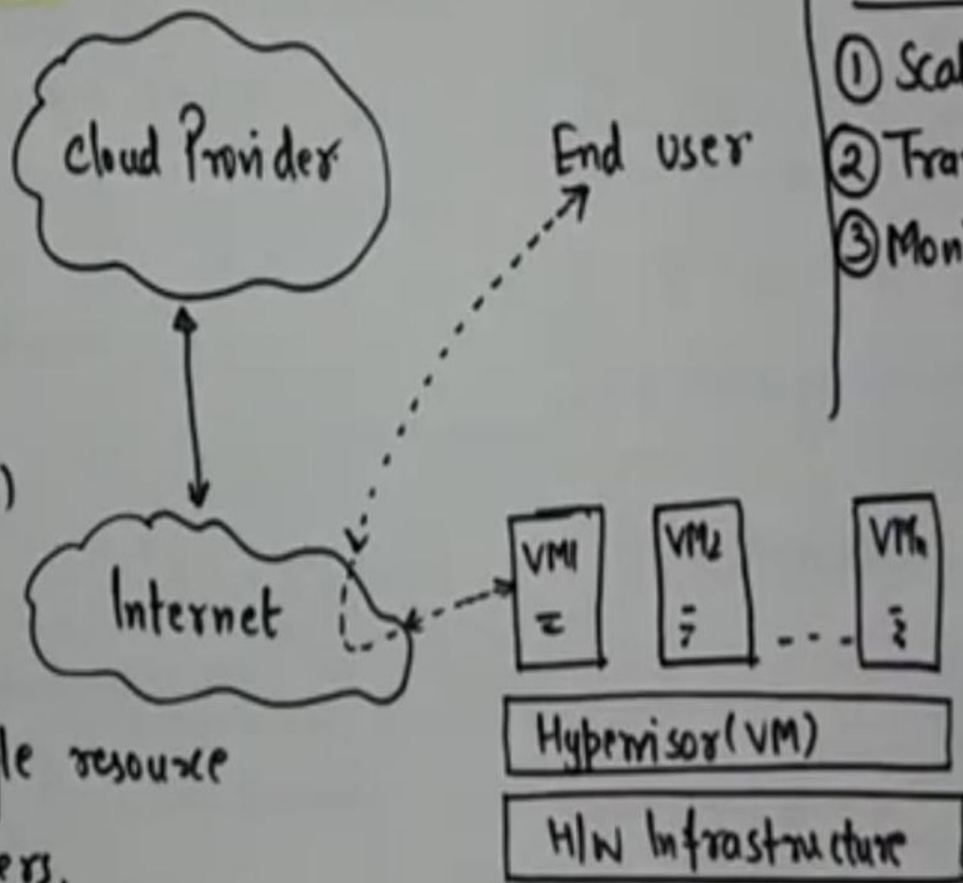
CLOUD COMPUTING Infrastructure:

- ↳ Storage devices
- ↳ Servers
- ↳ SW } cloud mgmt for users
- ↳ VM

Hypervisor:- firmware (low level)

- that acts as Virtual M/C managers.

→ Sharing of Single resource to multiple users.



Constraints:-

- ① Scalability
- ② Transparency.
- ③ Monitoring

Main Components of cloud Infrastructure:

- ↳ i) Front-end Interface for end-user.
- ii) Management for handling N/w resources
- iii) Storage for Virtual machine
- iv) Content Storage tool working in VM.
- v) Monitoring tools

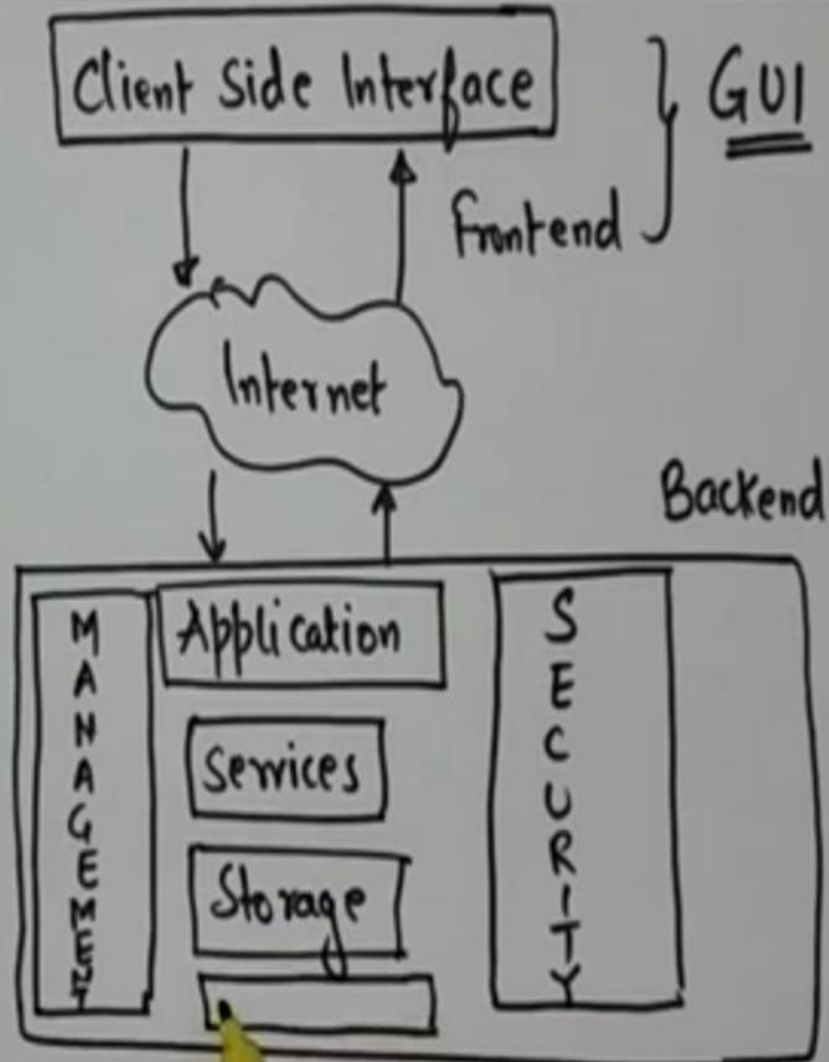
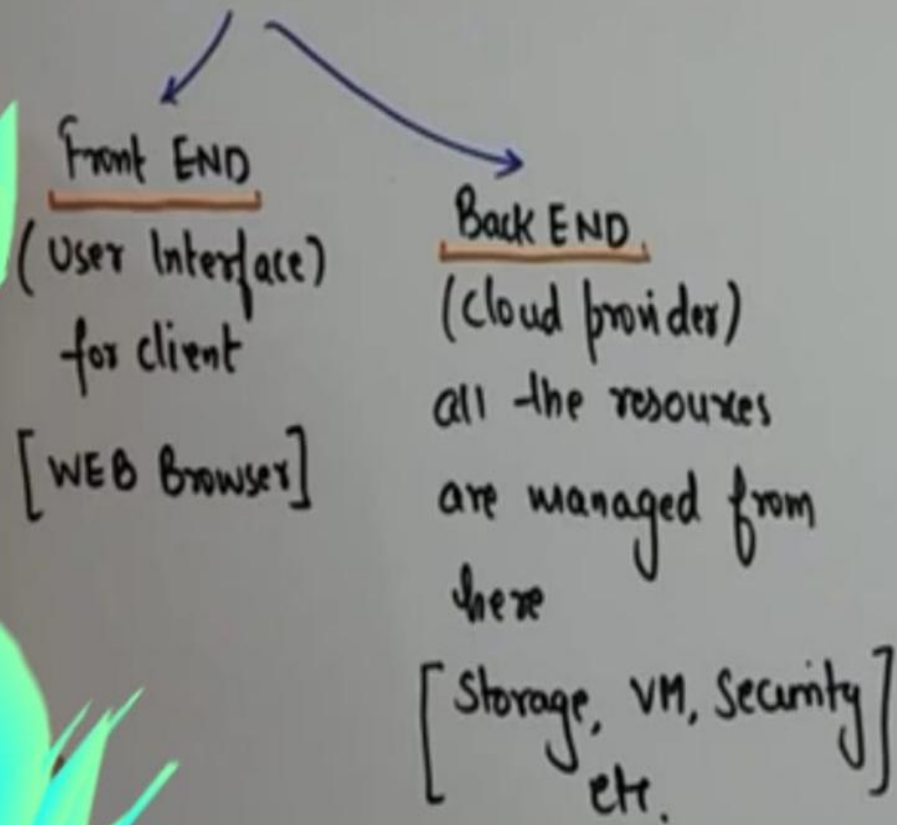
Types of Cloud Environment:

- ↳ i) End user has no idea about cloud complexity.
- ii) Cloud Service providers who has control over cloud environment.

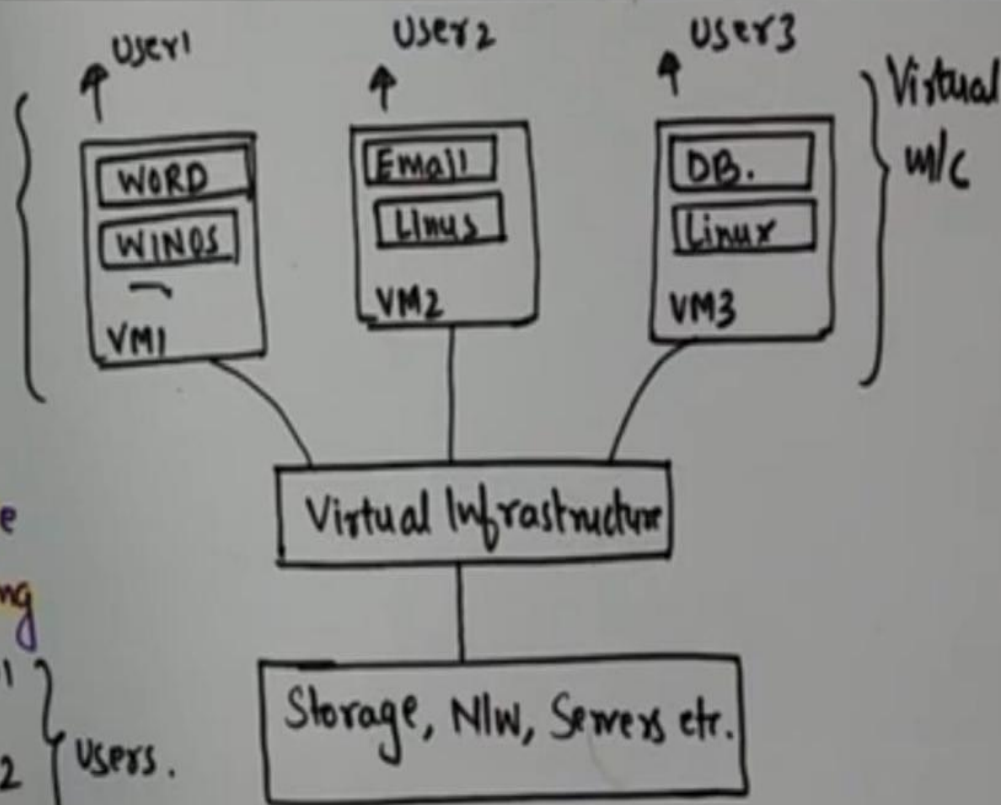
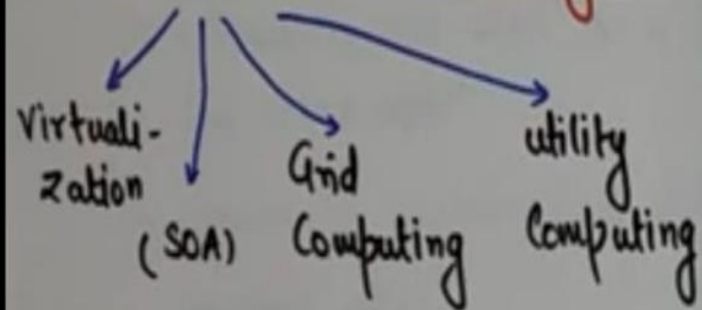
Limitations of cloud Computing:

- ↳ i) Availability of Services :- Accessed via Internet.
- ii) Data Lock-In :- Shifting of large vol. of data from one platform to other.
- iii) Data Segregation :- Isolation of data of each user.
- iv) Scaling Resources :- Sudden demand of increased resource come.
- v) Location of Data :- Geographically. Each country has its own rule.
- vi) Deletion of Data :- users demands complete removal of data.
- vii) Recovery and Backup :- How to recover

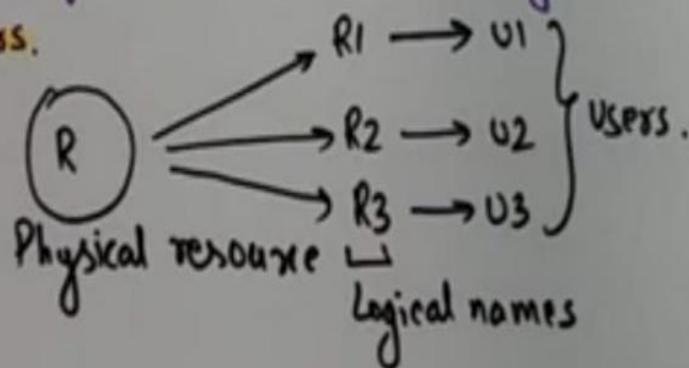
CLOUD COMPUTING - Architecture:



CLOUD COMPUTING- Technologies:-



(i) Virtualization: Allows to share single physical instance of an applⁿ / resource among multiple users.



Multitenant Architecture.

CLOUD COMPUTING APPLICATIONS:

(A) BUSINESS APPLICATIONS:

- i) Mail-chimp: E-mail Service [Marketing, Campaign]
- ii) Chatter- App: Share imp real time info in org.
- iii) Google Apps: [Business]
→ Spreadsheets, PPT, Doc file.
- iv) Quickbooks: e-accounting solⁿ.

(B) DATA STORAGE and BACKUP:

- i) Mozy: Backup Service [Data Loss prevention]
- ii) Box. com: interactive [Drag and Drop of files]
- iii) Joukuv: Google Docs] Single list of files.
Dropbox

(C) MANAGEMENT APPLICATIONS:

- i) Evernote: Organize sticky notes.
- ii) Togg1: Real time tracking of project
- iii) Outright: Track and manage financial data.

(D) SOCIAL APPLICATIONS:

Facebook: Social Nlw website

Twitter: Interact with Public.

CLOUD COMPUTING CHALLENGES:

Cloud Computing is used for enabling global access to mutual pool of resources such as services, apps, data, servers and computer N/w.

Challenges:

- i) COST: Tuning the Platform, Data transfer,
Maintenance, management } ↑ COST of cloud Computing.
- ii) RELIABILITY: ○ ← Jan
 ← X March } Service must be available when need.

- iii) DOWNTIME: To give better service downtime should be as low as possible.
- iv) SECURITY: Is your data secure.
- ↳ More security in passwords.
 - ↳ Access rights.
- v) DATA PRIVACY: (D) ← admin ← IT service
- All the info related to stored data must be kept private.
- vi) LOCK-IN: Agreement b/w user (and vendor).
- to make it consistent

CLOUD COMPUTING OPERATIONS:

Operations refers to delivering superior cloud service. } process of managing and delivering cloud services.

Allows to perform business ops

Online.

Ways to Manage cloud operations:

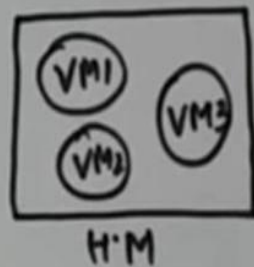
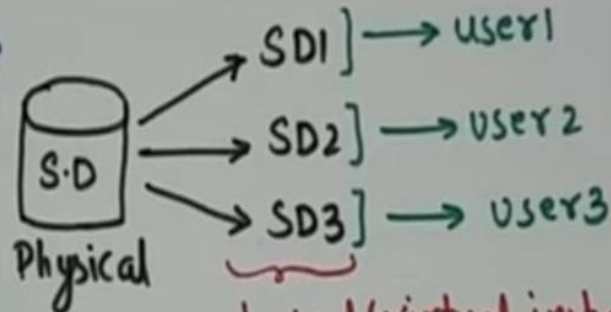
- ↳ Use of right tool to perform cloud func".
- ↳ doing things at right time
- ↳ doing things with right cost
- ↳ Standard to do work.

operations:-

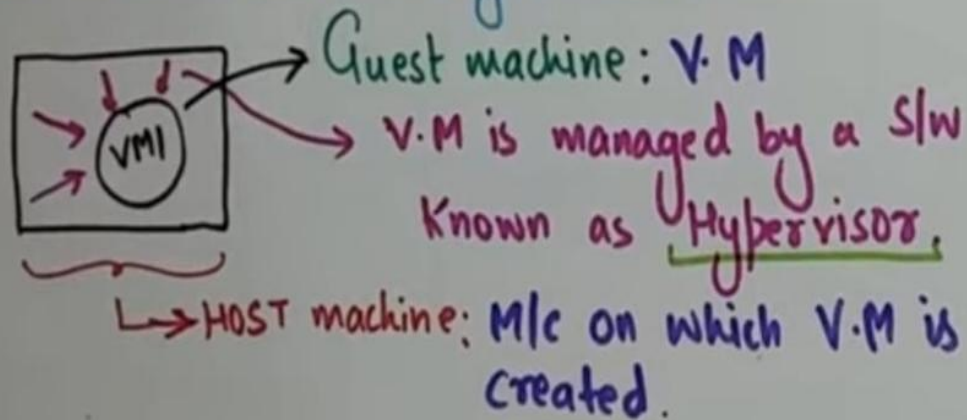
- ↳ E-mail marketing (Promotion)
- ↳ Content management
- ↳ Reporting
- ↳ marketplace
- ↳ Accounting Services
- ↳ HR Services

Cloud Computing Virtualization: Physical resources available as virtual resources.

Virtualization :- Allows to Share Single physical instance of an applⁿ or resource among multiple organizations.



Hardware Virtualization: It means creating Virtual machine over existing O.S and H/W.



Characteristics:-

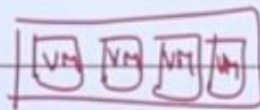
- ↳ ① Partitioning: Separate diff. V.M on single physical system.
- ② Isolation: Host and V.M should be isolated.
- ③ Encapsulation: V.M should be encapsulated in single file.

Type 1 virtualization

Bare Metal Hypervisor

A Type 1 (Bare-Metal Virtualization) hypervisor runs directly on the host's hardware, without needing an underlying operating system.

VMware ESXi, Microsoft Hyper-V,
Xen Hypervisor, KVM



Hypervisor

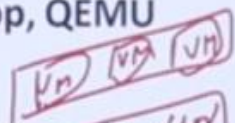
H/W CPU, RAM
SS, NIC, etc.

Type 2 virtualization

Hosted Hypervisor

A Type 2 (Hosted Virtualization) hypervisor runs on top of a host operating system (OS) rather than directly on the hardware.

VMware Workstation, Oracle VirtualBox,
Parallels Desktop, QEMU



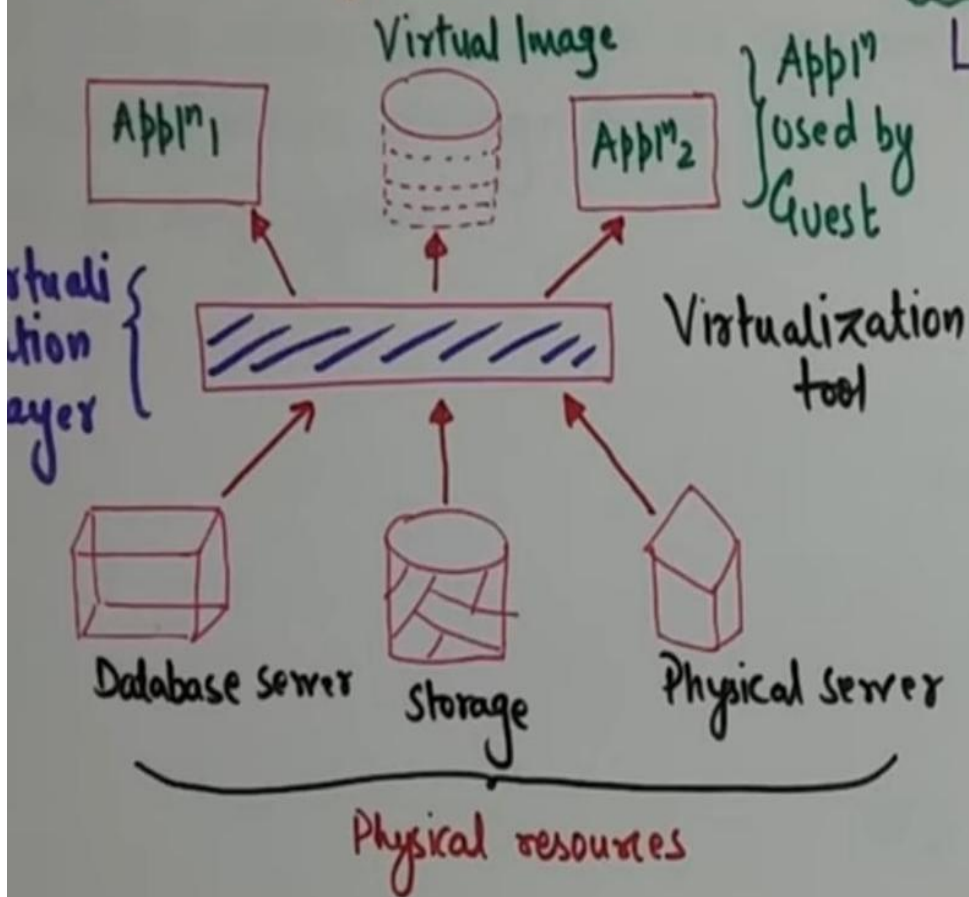
Hypervisor

Host OS

H/W

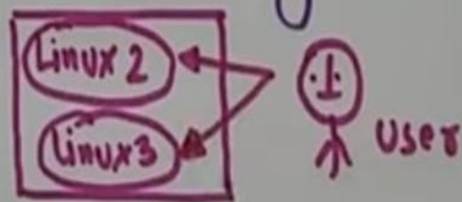
Feature	Type 1 (Bare-Metal)	Type 2 (Hosted)
Performance	High performance with direct hardware access ✓	Lower performance, depends on host OS
Setup Complexity	More complex, often used in data centers ✓ Ent M. ✓	Simple <u>setup</u> , ideal for <u>personal</u> use
Use Cases	Cloud providers, enterprise data centers, HPC	Development, testing, education
Ideal Environment	Production servers with dedicated hardware	Personal desktops and development setups

Virtualization Reference Model:



Advantages:-

- i) Allows access of n/w devices over any network.
- ii) Isolation of One App^m
- iii) User Can operate on diff. versions of the OS.
- iv) Optimization of Single System use.
- v) ↑ Reliability.



Types of Virtualization:

1. Hardware Virtualization.
2. Operating system Virtualization.
3. Server Virtualization.
4. Storage Virtualization.

1) Hardware Virtualization

When the virtual machine software or virtual machine manager (VMM) *is directly installed on the hardware system* is known as hardware virtualization.

The main job of hypervisor is to control and monitoring the processor, memory and other hardware resources.

After virtualization of hardware system we can install different operating system on it and run different applications on those OS.

1) Hardware Virtualization

When the virtual machine software or virtual machine manager (VMM) *is directly installed on the hardware system* is known as hardware virtualization.

The main job of hypervisor is to control and monitoring the processor, memory and other hardware resources.

After virtualization of hardware system we can install different operating system on it and run different applications on those OS.

Hardware virtualization is mainly done for the server platforms, because controlling virtual machines is much easier than controlling a physical server.

2) Operating System Virtualization:

When the virtual machine software or virtual machine manager (VMM) *is installed on the Host operating system* instead of directly on the hardware system is known as operating system virtualization.

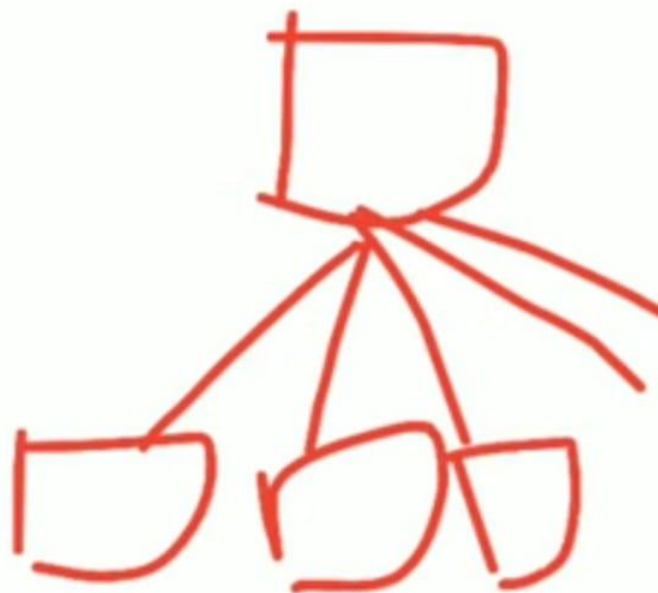
Operating System Virtualization is mainly used for testing the applications on different platforms of OS.

3) Server Virtualization:

When the virtual machine software or virtual machine manager (VMM) is *directly installed on the Server system* is known as server virtualization.

Usage:

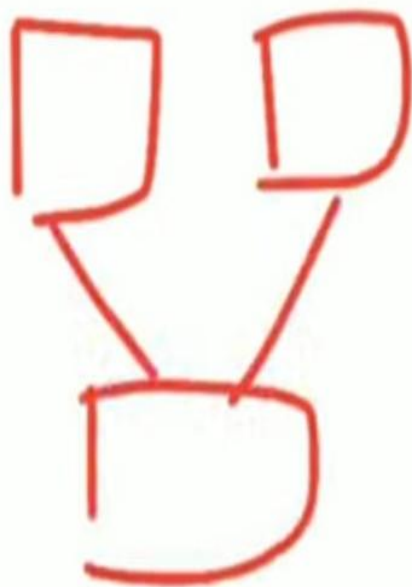
Server virtualization is done because a single physical server can be divided into multiple servers on the demand basis and for balancing the load.



4) Storage Virtualization:

Storage virtualization is the *process of grouping the physical storage from multiple network storage devices so that it looks like a single storage device.*

Storage virtualization is also implemented by using software applications.



Usage:

Storage virtualization is mainly done for back-up and recovery purposes.

What is Virtualization?



Virtualization

- **Virtualization** is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources".
- It allows to share a single physical instance of a resource or an application among multiple customers and organizations.
- A Virtual machine provides an environment that is logically separated from the underlying hardware.

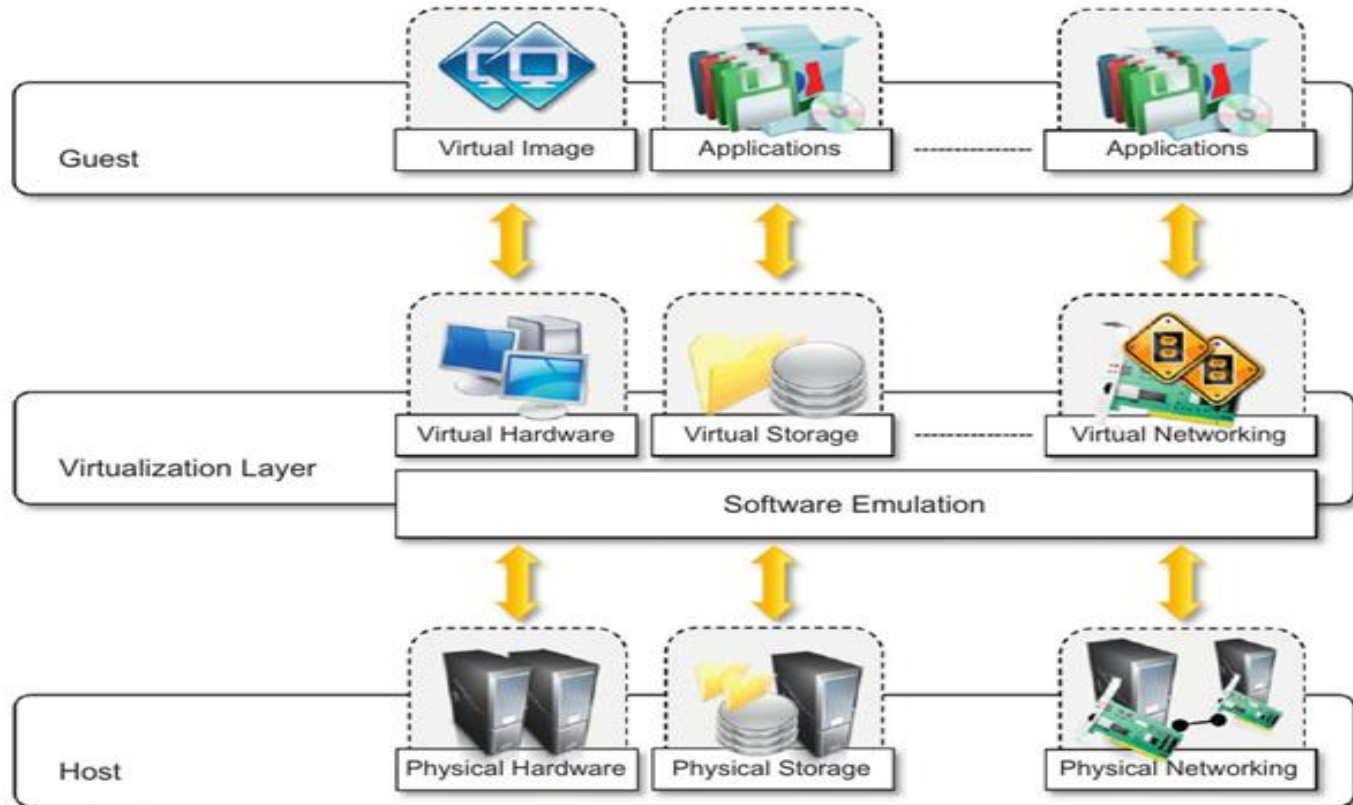
More about Virtualization

- Virtualization is a solution to optimize the resources available in the IT infrastructure.
- In general it is a creation of virtual machines and manage them from one place.
- A virtualization system partitions a single physical “machine” into multiple virtual machines and also decouples the OS from the dependence of hardware.
- The machine on which the virtual machine is going to create is known as **Host Machine** and that virtual machine is referred as a **Guest Machine**.

Virtualized Environments

- Three major components of Virtualized Environments
- Host — the original environment (the machine or software) on which the VM is implemented
- Guest — the OS that interacts with Virtualization Layer under the control of the VM
- Virtualization Layer (Hypervisor)— recreate the same or different environment where guest will run.

Virtualization Reference Model



Characteristics of Virtualization

- **1. Increased Security** – The ability to control the execution of a guest program in a completely transparent manner opens new possibilities for delivering a secure, controlled execution environment.
- A virtual machine manager can control and filter the activity of the guest programs, thus preventing some harmful operations from being performed.
- **2.Sharing** - Virtualization allows the creation of a separate computing environment within the same host.

Characteristics of Virtualization

- **3. Aggregation** – It is possible to share physical resources among several guests, but virtualization also allows aggregation, which is the opposite process. A group of separate hosts can be tied together and represented to guests as a single virtual host.
- **4. Emulation** -Guest programs are executed within an environment that is controlled by the virtualization layer, which ultimately is a program.
- Also, a completely different environment with respect to the host can be emulated, thus allowing the execution of guest programs requiring specific characteristics that are not present in the physical host.

Characteristics of Virtualization

- **6. Isolation –**
- Virtualization allows providing guests—whether they are operating systems, applications, or other entities—with a completely separate environment, in which they are executed.
- The guest program performs its activity by interacting with an abstraction layer, which provides access to the underlying resources.
- **7. Portability -** The concept of portability applies in different ways according to the specific type of virtualization considered.

Benefits of Virtualization

- Better resource utilization
- It helps in saving lots of cost required for IT infrastructure.
- It allows multiple operating systems on one virtualization platform.
- Allows remote access
- It removes the dependency of heavy hardware to run the application.
- Pay per use of IT infrastructure on demand

Types of Virtualization

Application Virtualization

- It helps a user to have remote access of an application from a server.
- The server stores all personal information and other characteristics of the application but can still run on a local workstation through the internet.
- Example of this technology are Citrix XenApp, Turbo (software), Symantec Workspace Virtualization etc.
- Application virtualization allows applications to run in environments that do not suit the native application.
- For example, Wine allows some Microsoft Windows applications to run on Linux.

Types of Virtualization

Desktop Virtualization

- allows the users' OS to be remotely stored on a server in the data center.
- It allows the user to access their desktop virtually, from any location by a different machine.
- Users who want specific operating systems other than Windows Server will need to have a virtual desktop.
- Main benefits of desktop virtualization are user mobility, portability, easy management of software installation, updates, and patches.
- Examples: Citrix XenDesktop, Wyse ThinOS, Microsoft Remote Desktop Services, Microsoft Enterprise Desktop Virtualization (MED-V);, VMware View Manager.

Types of Virtualization

Storage Virtualization

- Perhaps the easiest and cost-effective type of virtualization to implement
- storage virtualization involves assembling physical hard drives into a single application and pushing to the internet for virtual access.
- It is through this virtualization that cloud storage platforms are made possible.
- It helps businesses store data securely and also has backup and data recovery features.
- Dropbox. Google Drive, etc., are examples of cloud storage services.

Types of Virtualization

Server Virtualization

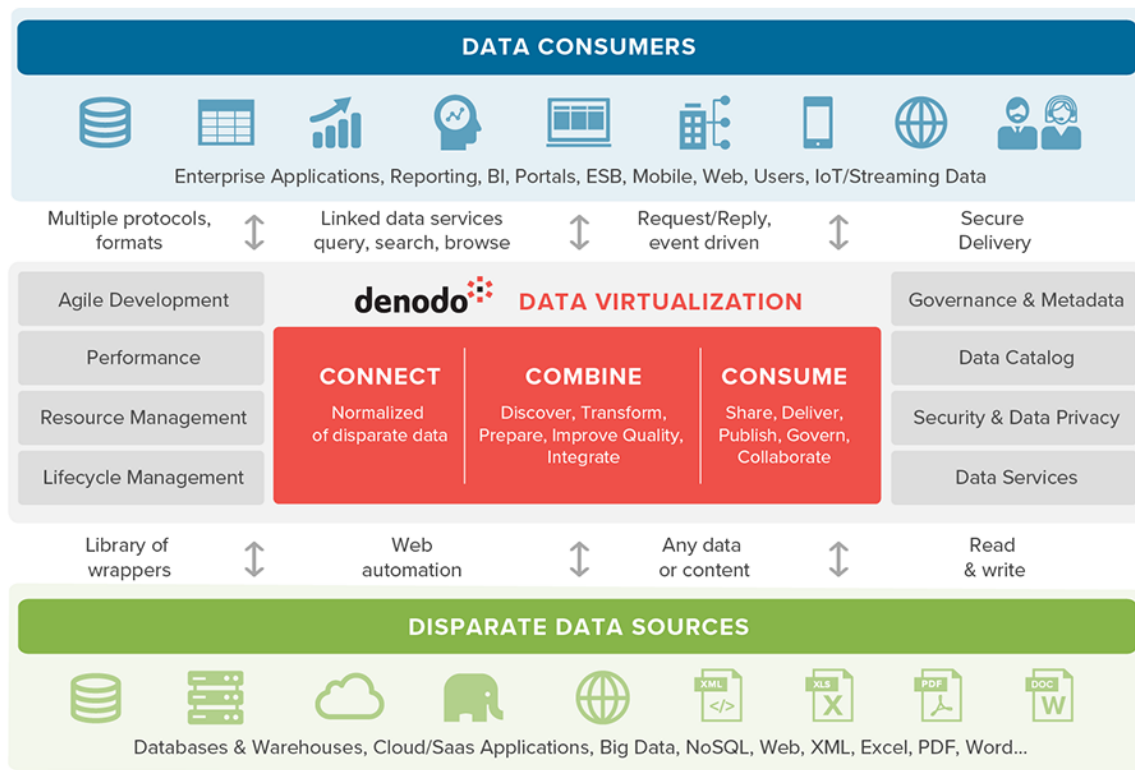
- Like other types of virtualization, developed servers are moved to the virtual environment to serve multiple users.
- The process involves dividing a physical server into several distinctive and secluded (hidden) virtual servers.
- This is usually done by using a software application.
- After which each virtual server independently runs its own operating system.
- This helps eliminate server complexity, increase application performance and also deploy workload speedily.
- Vendors – VMWare, Microsoft, Citrix, Amzon, Google etc

Types of Virtualization

Data Virtualization

- This is the kind of virtualization in which the data is collected from various sources and managed at a single place without knowing more about the technical information like how data is collected, stored & formatted then arranged that data logically so that its virtual view can be accessed by its interested people and stakeholders, and users through the various cloud services remotely.
- Many big giant companies are providing their services like Oracle, IBM, At scale, Cdata, etc.

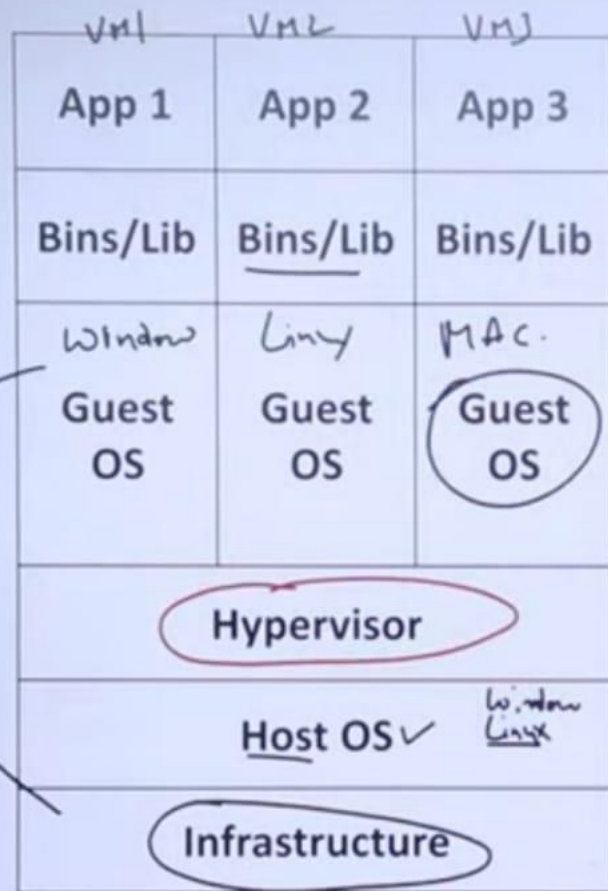
Types of Virtualization



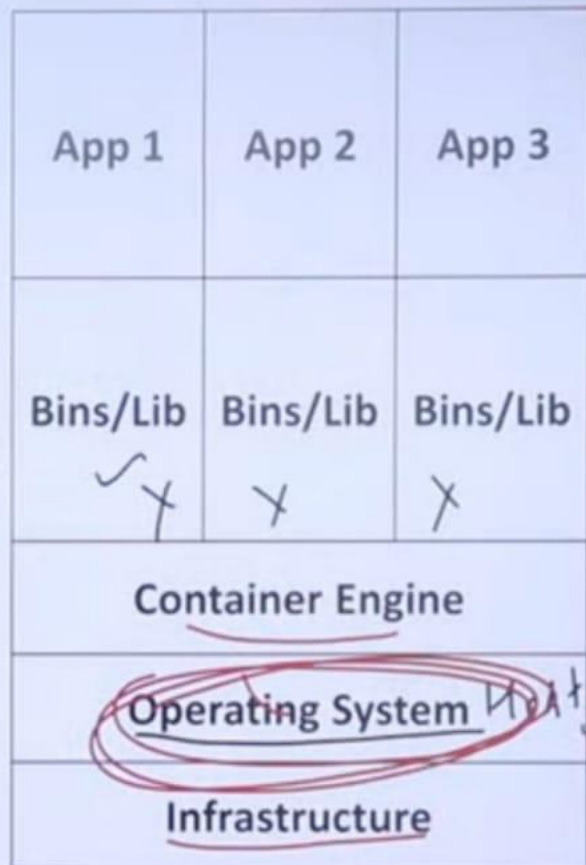
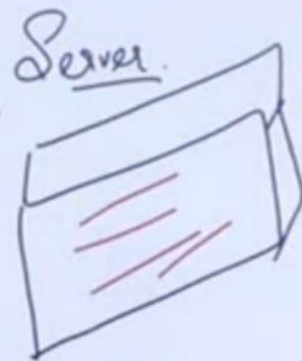
Types of Virtualization

Network Virtualization

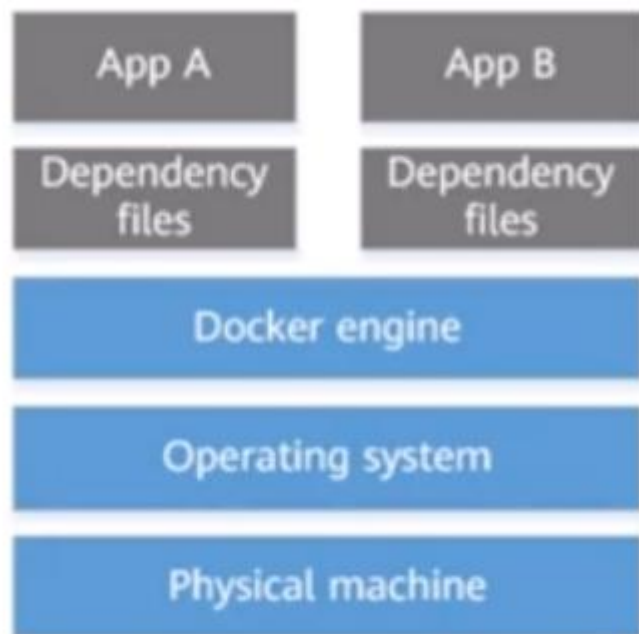
- is a process of logically grouping physical networks and making them operate as single or multiple independent networks called Virtual Networks.
- Network Virtualization Platform: VMware NSX.



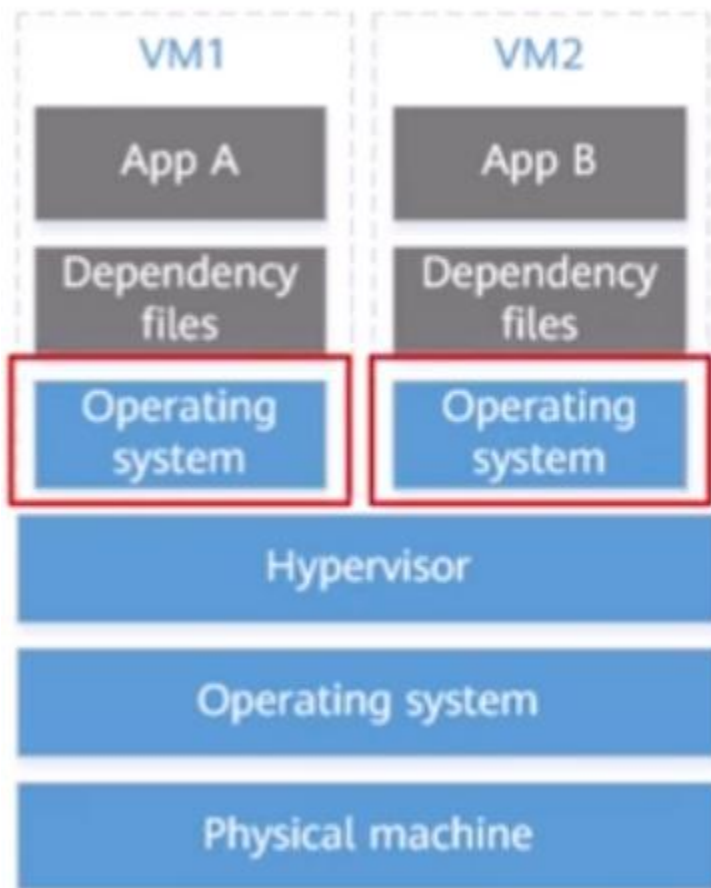
Virtualization



Containerization

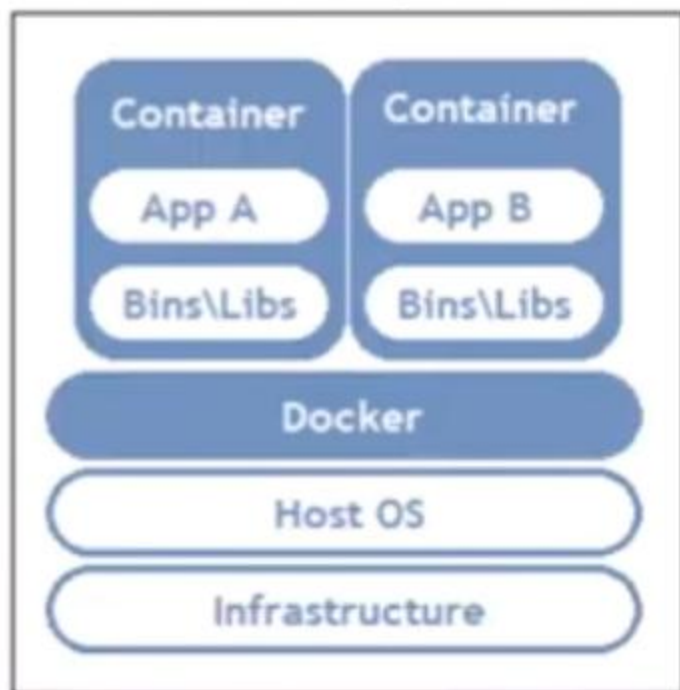


Docker container



VM

Container Based Implementation



Data Centers are becoming Warmer

- Modern data centers in the cloud computing environment host a variety of applications that run from few seconds to longer periods.
- Because of insufficient hardware usage and inefficient resource usage, energy costs have increased.
- Consequently, energy cost for operating and cooling the data centers will be more than hardware acquisition cost.

VM Consolidation

- VM Consolidation is a technique to reduce the number of active physical machines by migrating and consolidating VMs on active physical machines.
- Consolidation includes two strategies:
- VM Placement
- VM Migration

VM Placement

- Virtual machine placement is a process of mapping a group of virtual machines (VMs) onto a set of physical machines (PMs) in a data center.
- The aim is to maximize resource utilization and minimize the total power consumption by PMs.
- It can be done either by saving energy by shutting down some servers or it can be maximizing the resource utilization.

VM Migration

- VM Migration is carried out after initial VM Placement in order to reduce the number of running physical machines by migration of few VMs and consolidate into reduced number of PMs.

VM Migration : Types

- There are two types of VM migration:
- cold and live.
- Cold migration occurs when the VM is shut down.
- Live migration occurs while the VM is actually running.

VM Migration

- Depending on the virtual machine resource type, you can perform three types of migration.
- **Change compute resource only**
- Moving a virtual machine, but not its storage, to another compute resource, such as a host, cluster, resource pool.
- **Change storage only**
- Moving a virtual machine and its storage, including virtual disks, configuration files, or a combination of these, to a new datastore on the same host.
- **Change both compute resource and storage**
- Moving a virtual machine to another host and at the same time moving its disk or virtual machine folder to another datastore.

Steps involved in VM migration process

- Select the PM which is overload or under load.
- Select one or more VMs from under loaded server
- Select the appropriate destination PM where selected VMs can be placed and
- Transfer / migrate the VMs to destination PM.

Types of Hypervisor

Two types of Hypervisor

Type 1 Hypervisor - Also known as native or bare metal hypervisor

- It runs directly on hardware to manage virtual machines

Type 2 Hypervisor- known as hosted hypervisor

- It is installed on operating System.

Virtualization Techniques

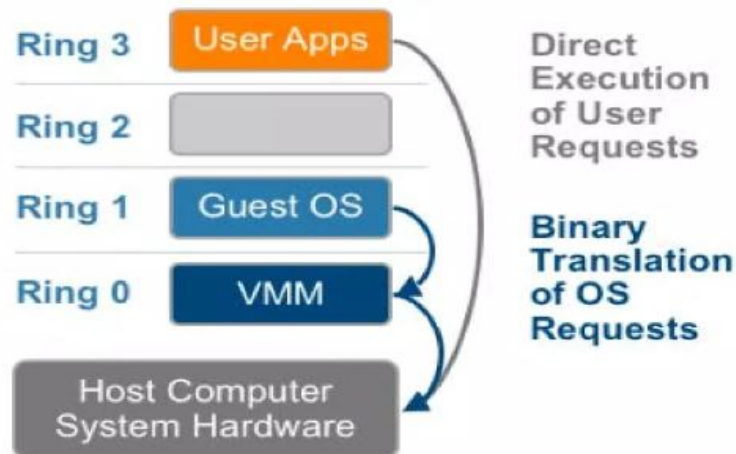
There are multiple approaches to running the guest operating system. These include:

- **Full Virtualization**
- **Paravirtualization**
 - Also known as OS assisted virtualization
- **Hardware-assisted virtualization**
 - Also known as accelerated virtualization, hardware virtual machine (HVM)

Virtualization Techniques

Full Virtualization

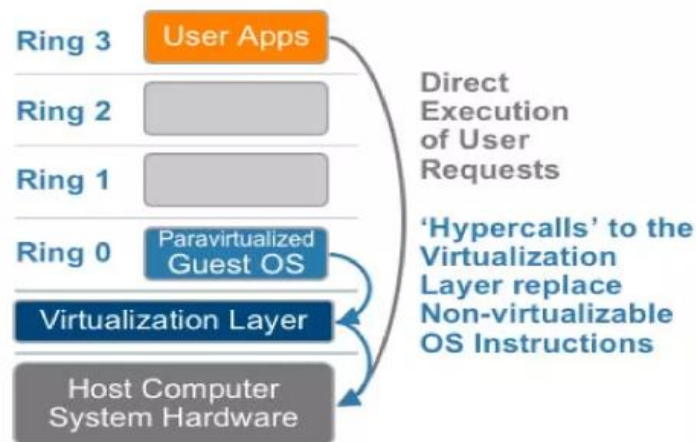
- Completely abstracted from the underlying hardware by virtualization layer
- Guest OS unaware that it is a guest
- Hypervisor translates all OS calls on-the-fly
- No hardware assistance or modification; flexibility



Virtualization Techniques

Paravirtualization

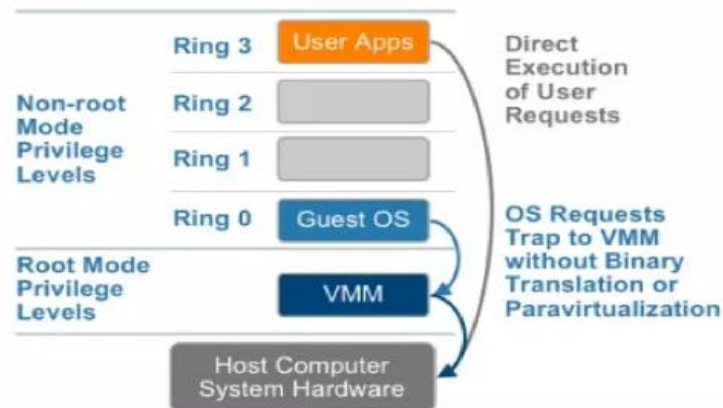
- An efficient and lightweight virtualization technique
- The hypervisor provides an API and the Guest OS calls that API, requiring OS modifications.
- Does not require virtualization extensions from the host CPU
- Guests and control domains require kernel support and drivers
- Enable near-native performance
- *paravirt-ops* code supported by Linux kernel as of version 2.6.23



Virtualization Techniques

Hardware-assisted virtualization

- Enables efficient full virtualization using help from hardware capabilities, primarily from the host processors
- Unmodified guest OS; no API calls made
- Hypervisor traps sensitive calls
- Added to x86 processors (Intel VT-x or AMD-V) in 2006



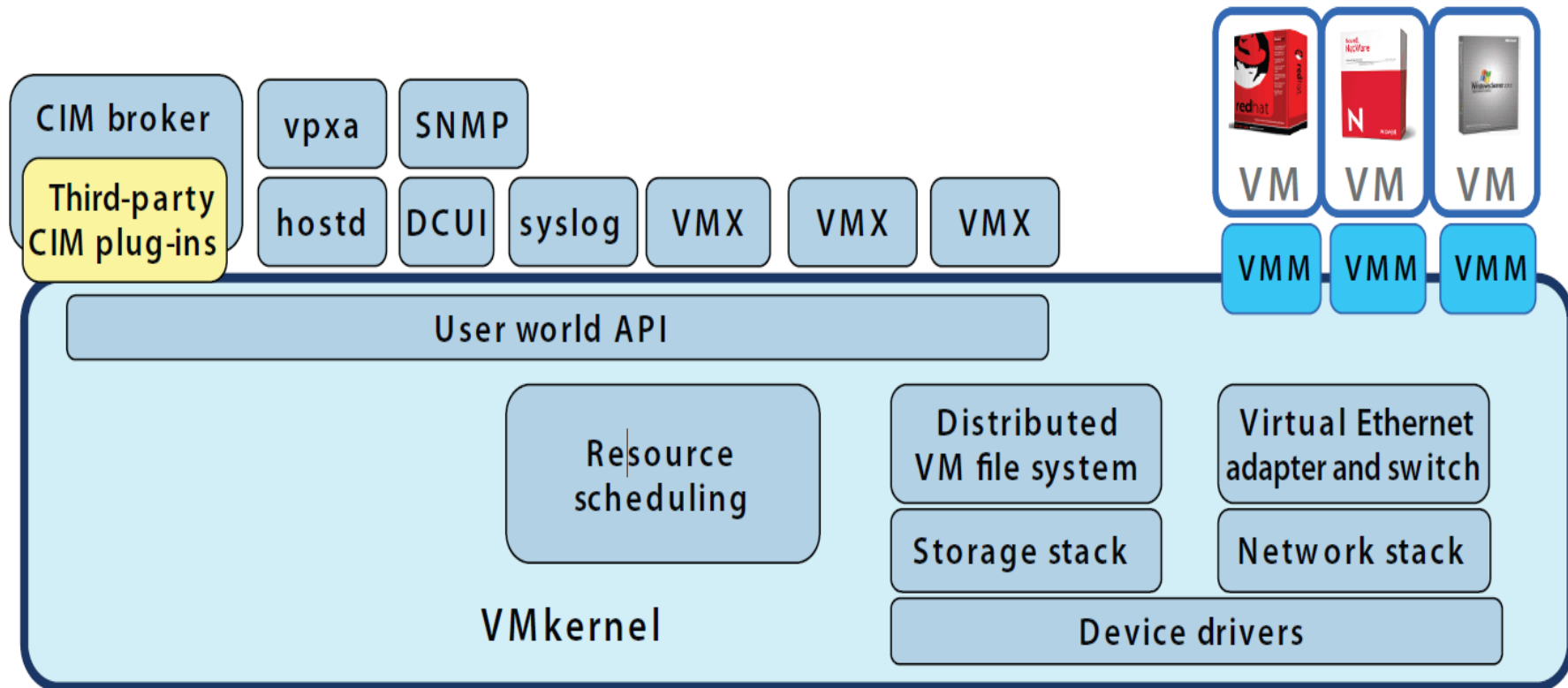
VMWare ESXi

- The VMware ESXi architecture comprises the underlying operating system, called VMkernel, and processes that run on top of it.
- VMkernel provides means for running all processes on the system, including management applications and agents as well as virtual machines.
- It has control of all hardware devices on the server, and manages resources for the applications.
- The main processes that run on top of VMkernel are:
 - Direct Console User Interface (DCUI) — the low-level configuration and management interface, accessible through the console of the server, used primarily for initial basic configuration.

VMWare ESXi

- **The virtual machine monitor**, which is the process that provides the execution environment for a virtual machine, as well as a helper process known as VMX.
- Each running virtual machine has its own VMM and VMX process.
- Various agents used to enable high-level VMware Infrastructure management from remote applications.
- The Common Information Model (CIM) system: CIM is the interface that enables hardware-level management from remote applications via a set of standard APIs.

VMWare ESXi Architecture



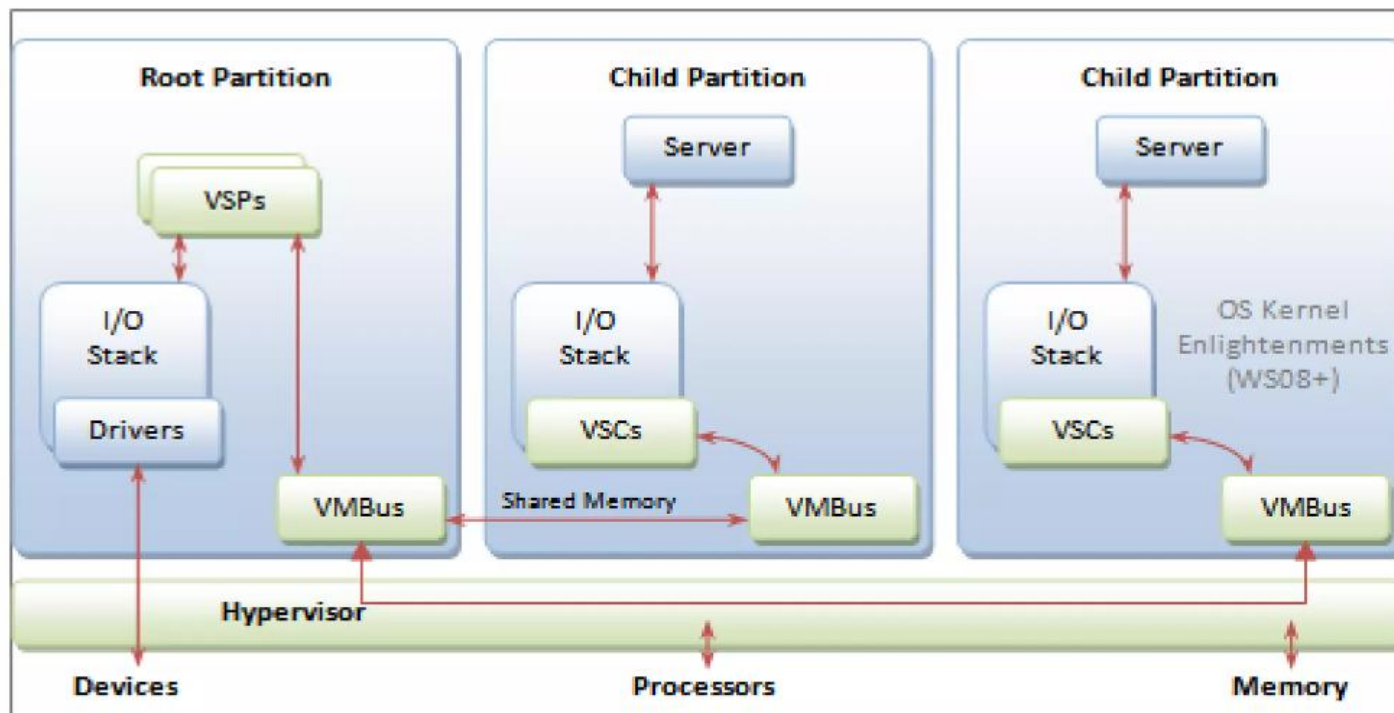
Microsoft Hyper-V Architecture

- Hyper-V features a Type 1 hypervisor-based architecture.
- The hypervisor virtualizes processors and memory and provides mechanisms for the virtualization stack in the root partition to manage child partitions (virtual machines) and expose services such as I/O devices to the virtual machines.

Microsoft Hyper-V Architecture

- The root partition owns and has direct access to the physical I/O devices.
- The virtualization stack in the root partition provides a memory manager for virtual machines, management APIs, and virtualized I/O devices.
- It also implements emulated devices such as the integrated device electronics (IDE) disk controller and PS/2 input device port, and it supports Hyper-V-specific synthetic devices for increased performance and reduced overhead

Microsoft Hyper-V Architecture



Microsoft Hyper-V Architecture

- The Hyper-V-specific I/O architecture consists of virtualization service providers (VSPs) in the root partition and virtualization service clients (VSCs) in the child partition.
- Each service is exposed as a device over VMBus, which acts as an I/O bus and enables high-performance communication between virtual machines that use mechanisms such as shared memory.
- The guest operating system's Plug and Play manager enumerates these devices, including VMBus, and loads the appropriate device drivers (virtual service clients).

